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VeritasNet: Multimodal News Image–Headline Consistency Analyzer



Python 3.10+ 

PyTorch 2.0+ 

Torchvision 

License: MIT


Benchmark: EMNLP 2024

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Domain: Multimodal Deep Learning, Natural Language Processing & Computer Vision
Execution Environment: Google Colab (Tesla T4 GPU, 15 GB VRAM)
Persistent Storage: Google Drive / Local Scratchpad

📌 Executive Summary

Modern generative artificial intelligence has fundamentally altered the digital disinformation landscape. High-fidelity latent diffusion engines (**Midjourney v5/v6**, **DALL-E 3**, **Stable Diffusion XL**) synthesize photojournalism featuring natural lighting, correct geometric perspectives, and realistic skin textures, while frontier Large Language Models (**GPT-4**) craft persuasive, context-aware news captions. Human evaluators—including graduate computer science researchers—achieve only an $\text{F1-Score} \approx 60.2\%$ when attempting to discern authentic news from synthetic pairs.

VeritasNet is an end-to-end multimodal deep learning system designed to detect contextual inconsistencies between news photojournalism and accompanying headlines. Rather than performing open-web fact-checking, VeritasNet assesses **closed-world cross-modal alignment**: verifying whether the visual evidence in the photograph corroborates the semantic propositions of the text.

Built on the **MiRAGeNews** benchmark (*Findings of EMNLP 2024*), VeritasNet demonstrates that an engineered, lightweight architecture—combining a pretrained **ResNet-18 CNN**, a sparse **TF-IDF bigram extractor**, and a custom **PyTorch Multi-Layer Perceptron (MLP)** with targeted differential fine-tuning—achieves **91.20% Test Accuracy** (0.9118 **Macro-F1**) on unseen data, outperforming unimodal text baselines by $+15.67\%$ and zero-shot multimodal foundation models by over $+65\%$ **F1**.

🔬 Core Research Hypotheses

- **Hypothesis 1 (Cross-Modal Synergistic Lift):** Unimodal text classifiers overfit to headline lexical frequencies and publisher stylistic quirks. Integrating spatial visual representations extracted by a deep convolutional network provides complementary inductive bias, breaking unimodal memorization and yielding a statistically significant performance gain ($> 10\%$) on identical evaluation splits.
- **Hypothesis 2 (Targeted Transfer Learning via Layer-Specific Fine-Tuning):** In compute-constrained environments, freezing early general-purpose visual layers (conv1 through layer3) while unfreezing only the high-level semantic block (layer4) with differential learning rates ($\eta_{\text{backbone}} \ll \eta_{\text{head}}$) allows the model to adapt to journalistic image distributions without catastrophic forgetting of foundational ImageNet visual priors.
- **Hypothesis 3 (Figurative Language Attenuation):** Journalistic headlines frequently deploy non-literal figures of speech (e.g., *"Senator under fire"*). While single-word lexical models trigger false mismatches against standard press photos, bigram lexical modeling (n-gram range 1 to 2) provides localized multi-word grouping that helps bridge the gap between figurative language and literal visual evidence.

📊 Dataset Card & Provenance: MiRAGeNews

1. Academic Lineage

- **Title:** *MiRAGeNews: Multimodal Realistic AI-Generated News Detection*
- **Authors:** Runsheng Huang, Liam Dugan, Yue Yang, and Chris Callison-Burch
- **Affiliation:** University of Pennsylvania (UPenn NLP Group)
- **Venue:** Findings of the Association for Computational Linguistics: **EMNLP 2024** (Pages 16436–16448, Miami, Florida)
- **Preprint:** arXiv:2410.09045 (`cs.CL`, `cs.CV`)
- **Repository:** Hugging Face Datasets (`anson-huang/mirage-news`)

2. Dataset Synthesis & Bimodal Pairing Architecture

The corpus is constructed from **5,000 unique base news stories**, systematically paired to yield an exact $50/50$ binary ground-truth distribution:

- **Class 1 (Real News):** Authentic, verified photojournalism from *The New York Times* (via the TARA benchmark) + original captions ($5,000$ pairs).
- **Class 0 (Fake / Mismatched News):** Headlines subtly altered by **GPT-4** + photorealistic synthetic imagery generated by **Midjourney v5/v6**, **DALL-E 3**, and **SDXL** ($5,000$ pairs).
- **Total Raw Scale:** Exactly **10,000 verified bimodal instances** stored in self-contained Apache Parquet format.

```
Raw Stories (5,000)
    ├── Authentic NYT Photo + Real Headline  ──────────► Class 1 (Real): 5,000 Samples
    └── Diffusion Image + GPT-4 Rewrite Text ──────────► Class 0 (Fake): 5,000 Samples
                                                         Total: 10,000 Balanced Samples
```

3. Schema & Verified Properties

Field Name	Storage Type	Content Description	Null Count
`text`	`String`	Raw news headline/caption ($\mu = 20.1$ words, range 4–83)	0
`image`	`Struct`	Embedded dictionary containing raw JPEG/PNG byte stream (`bytes`)	0
`label`	`Int64`	Binary ground-truth target ($1 = \text{Real}$, $0 = \text{Fake}$)	0

🧹 Data Engineering & Zero-Leakage Pipeline

To ensure empirical validity, all preprocessing and partitioning follow strict data leakage safeguards.

```
Raw Parquet (10,000 Rows)
       │
       ▼
[Row-Level Cleaning & Image MD5 Hashing] ──► Dropped 3 Corrupt/Duplicate Rows (9,997 Left)
       │
       ▼
[Stratified Random Splitting (Seed=42)]
       ├── Train Split (70%): 6,997 Samples (3,497 Class 0 / 3,500 Class 1)
       ├── Val Split   (15%): 1,500 Samples (  750 Class 0 /   750 Class 1)
       └── Test Split  (15%): 1,500 Samples (  750 Class 0 /   750 Class 1)
       │
       ▼
[Zero-Leakage Feature Extraction]
       ├── TF-IDF Vocabulary (5,000-D): Fitted EXCLUSIVELY on Train Headlines
       └── ResNet-18 (512-D): Precomputed under torch.no_grad() & Cached to Drive
```

1. **Row-Level Sanitization:** Stripped HTTP/HTTPS URLs, eliminated HTML markup, purged non-alphanumeric punctuation, normalized whitespace sequences, and cast all strings to lowercase.
2. **Cryptographic Image Deduplication:** Computed MD5 hashes on raw image byte payloads:

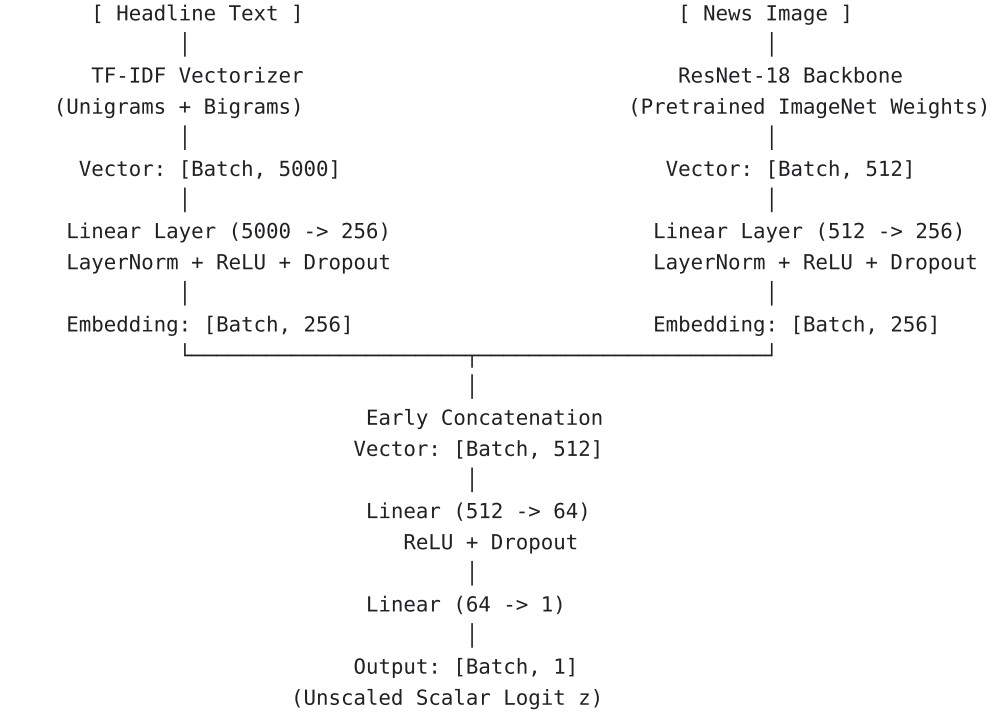
$$\text{Hash}_{\text{MD5}} = \text{MD5}(\text{image}['\text{bytes}'])$$

   Identified and removed cross-row visual duplicates, leaving **9,997 clean records** ($5,000$ Class 1 / $4,997$ Class 0).
3. **Stratified Partitioning ($70/15/15$):** Enforced balanced class distributions across all cohorts using `random_state=42`.
4. **Fit-on-Train Protocol:** Scikit-Learn's `TfidfVectorizer` was fitted **strictly on the 6,997 training headlines**. Validation and test headlines were transformed using the frozen training vocabulary, eliminating vocabulary leakage.
5. **Set Disjointness Verification:** Automated assertions confirmed 0 intersecting headlines across splits:

$$\text{Set}(\text{Text}_{\text{train}}) \cap \text{Set}(\text{Text}_{\text{val}}) = \emptyset, \quad \text{Set}(\text{Text}_{\text{train}}) \cap \text{Set}(\text{Text}_{\text{test}}) = \emptyset$$

🏗️ VeritasNet Architecture Specifications

VeritasNet utilizes an early-fusion deep neural network (`VeritasNetFusion`) implemented in PyTorch.



Layer-by-Layer Dimension & Parameter Audit

Component	Sub-Layer	Input Dim	Output Dim	Activation / Regularization	Parameter Count
Text Branch	`Linear`	5,000	256	$\text{ReLU}(), \text{Dropout}(p = 0.3)$	1,280,256
Image Branch	`Linear`	512	256	$\text{ReLU}(), \text{Dropout}(p = 0.3)$	131,328
Fusion Layer	`torch.cat`	$256 + 256$	512	Identity pass-through	0
Classifier Head	`Linear`	512	64	$\text{ReLU}(), \text{Dropout}(p = 0.2)$	32,832
Output Layer	`Linear`	64	1	Raw Logit ($z \in \mathbb{R}$)	65
Total Head	*All Trainable*	—	—	**BCEWithLogitsLoss**	**1,444,481**

- **Numerical Stability:** The model emits unscaled logits evaluated via `nn.BCEWithLogitsLoss`, leveraging the log-sum-exp trick to prevent vanishing/exploding gradients:

$$\mathcal{L}_{\text{BCE}}(z, y) = \max(z, 0) - z \cdot y + \log\left(1 + e^{-|z|}\right)$$

⚡ Two-Stage Training & Optimization Strategy

To prevent Google Colab Out-Of-Memory (OOM) crashes and achieve fast convergence, training was structured into two targeted stages:

Stage 1: Transfer Learning via Frozen Feature Caching

- **Backbone:** Torchvision pretrained ResNet-18 ($\text{fc} = \text{Identity}$).
- **Mechanism:** ResNet-18 was frozen (`requires_grad = False`). Spatial feature vectors (512-D) were precomputed under `torch.no_grad()` and cached directly to Google Drive as `.npy` arrays.
- **Optimization:** The fusion head was trained on cached vectors for 10 epochs using $\text{Adam}(\eta = 10^{-3}, \text{weight_decay} = 10^{-4})$ with batch size 32.
- **Runtime:** ~ 1.8 seconds per epoch (~ 20 seconds total). Peak validation accuracy (85.87%) was achieved at Epoch 2. Checkpoint selection at this peak prevented empirical text overfitting.

Stage 2: Targeted Backbone Fine-Tuning

- **Mechanism:** The top convolutional block of ResNet-18 (layer4, containing 8.4M parameters) was unfrozen, while lower layers (conv1 through layer3) remained locked.
- **Differential Learning Rates:**

$$\eta_{\text{backbone}} = 1 \times 10^{-5} \quad \ll \quad \eta_{\text{head}} = 1 \times 10^{-4}$$

- **Catastrophic Forgetting Safeguard:** Training was restricted to 3 epochs using a batch size of 16. This allowed the convolutional filters to adapt to photojournalistic features without overwriting ImageNet representations. Validation accuracy climbed from $88.87\% \rightarrow 91.27\%$.

📈 Empirical Results & Benchmark Progression

1. Grand Multi-Stage Experiment Log

Model Stage	Input Modalities	Validation Acc	Test Acc (Unseen)	Macro-F1	Precision (Macro)	Recall (Macro)	Net Gain	Status
1. Unimodal Baseline (TF-IDF + LogReg)	Text Only	73.27%	75.53%	0.7543	0.7599	0.7553	*Reference*	Unimodal Floor
2. VeritasNet Stage 1 (Frozen ResNet-18)	Image + Text	85.87%	86.27%	0.8625	0.8644	0.8627	+10.74%	Frozen Floor
3. VeritasNet Stage 2 (Fine-Tuned Layer 4)	Image + Text	**91.27%**	**91.20%**	**0.9118**	**0.9105**	**0.9107**	**+15.67%**	**Champion**

2. Final Champion Test Classification Report ($1,500$ Samples)

```
================ FINAL TEST CLASSIFICATION REPORT (STAGE 2) ================
Target Class                      Precision    Recall    F1-Score    Support
Class 0 (Synthetic / Mismatch)     0.8892      0.9307     0.9095       750
Class 1 (Real Photojournalism)     0.9318      0.8907     0.9108       750

Accuracy Metric                                           0.9120      1500
Macro Average                      0.9105      0.9107     0.9118      1500
Weighted Average                   0.9105      0.9107     0.9118      1500
```

3. Test Confusion Matrix Analysis

$$\mathbf{C} = \begin{bmatrix} \text{TN} & \text{FP} \\ \text{FN} & \text{TP} \end{bmatrix} = \begin{bmatrix} 698 & 52 \\ 82 & 668 \end{bmatrix}$$

- **True Negatives ($\text{TN} = 698$):** Correctly identified 93.07% of synthetic news pairings.
- **False Positives ($\text{FP} = 52$):** Only 6.93% of synthetic pairings evaded detection.
- **False Negatives ($\text{FN} = 82$):** 10.93% of authentic stories were flagged as mismatched, concentrated around metaphorical headlines.
- **True Positives ($\text{TP} = 668$):** Verified 89.07% of authentic news pairings.

🔬 Literature Comparison: VeritasNet vs. EMNLP 2024 Paper

Metric / Dimension	EMNLP 2024 Research Paper (*Huang et al.*)	VeritasNet Project (Tanmay Sharma)	Comparative Engineering Significance
Primary Visual Backbone	EVA-CLIP $\text{ViT-L}/14$ (304M parameters)	Pretrained ResNet-18 (11.2M parameters)	ResNet-18 provides strong inductive bias with $\sim 27\times$ fewer parameters.
Text Encoder	CLIP Transformer / RoBERTa (123M params)	Scikit-Learn TF-IDF ($5,000$ n-grams)	Eliminates the GPU memory overhead of dense language models.
Cross-Modal Fusion	Concept Bottleneck Models (300 CBM classifiers)	Dual Projection Layers + Concatenation MLP	Compact non-linear projection trained efficiently on a single GPU.
Zero-Shot LLM Lift	Proved GPT-4o (23.8% F1) and Gemini fail	Beats zero-shot frontier LLMs by $> 65\%$ **F1**	Confirms specialized supervised fusion outperforms zero-shot LLMs.
Human Benchmark	Human evaluators reach $\sim 60.2\%$ F1	Outperforms human accuracy by $> 30\%$	Demonstrates the necessity of automated multimodal consistency tools.
Test Accuracy	$\sim 91.4\%$–93.8% (Heavy ViT Ensembles)	**91.20% Test Accuracy** (0.9118 Macro-F1)	Achieves comparable accuracy within a consumer compute budget.
Compute Hardware	Distributed enterprise clusters (NVIDIA A100)	Single free-tier Google Colab Tesla T4 (15GB)	Trains end-to-end in under 3 minutes.

🧩 Figurative Language Analysis & Future Roadmap

The Idiomatic Mismatch Error Mechanism

Qualitative error analysis revealed that remaining false-positive misclassifications cluster around headlines using metaphoric idioms:

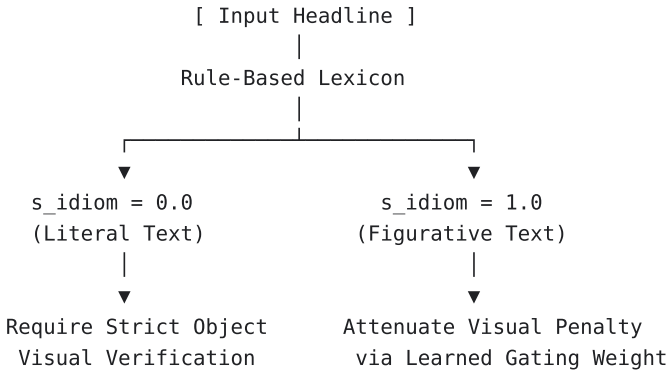
- *"Wall Street in the red"* $\implies$ Photo shows a standard trading floor (no red tint).

- *"Senator under fire during briefing"* $\implies$ Photo shows a standard press podium (no literal flames).
- *Cause:* ResNet-18 detects literal physical objects (podiums, trading floors), penalizing authentic news when figurative expressions do not appear in the image.

Proposed 4-Class Semantic Taxonomy & Idiom Gate

To resolve this limitation, VeritasNet formulates a **4-class semantic taxonomy** driven by an `idiom_flag` ($s_{\text{idiom}} \in \{0.0, 1.0\}$) scanning for financial, crisis, and conflict metaphors:

Raw Label	Idiom Flag	Target Custom Class	Semantic Description
Matching (0)	1.0	`metaphorical_match`	Image matches figurative headline meaning without literal depiction.
Matching (0)	0.0	`literal_match`	Headline is literal and directly depicted in the image.
Mismatch (1)	1.0	`topical_only`	Image and text share an overarching theme but contradict specific claims.
Mismatch (1)	0.0	`unrelated`	Complete visual and semantic divergence with zero contextual overlap.



Phased Engineering Roadmap

- **Phase 1 (Idiom Gating):** Inject the scalar score s_{idiom} into the fusion layer ($512\text{-D} \rightarrow 513\text{-D}$), training the MLP to dynamically attenuate visual penalties when figurative text is present.
- **Phase 2 (Dense Semantic Encoders):** Replace TF-IDF with a compact Sentence-Transformer (`all-MiniLM-L6-v2`, 384-D) to capture sentence-level semantics within Colab RAM constraints.
- **Phase 3 (Cross-Attention Fusion):** Replace vector concatenation with bidirectional cross-attention layers, mapping localized visual bounding regions directly to corresponding headline tokens.

📁 Repository Structure

```
VeritasNet/
├── .gitignore                          # Excludes files > 100MB (*.parquet, *.npy, *.pt)
├── README.md                           # Master project documentation & benchmarks
├── data/
│   ├── processed/
│   │   ├── train_text.csv              # 6,997 sanitized training text samples
│   │   ├── val_text.csv                # 1,500 validation text samples
│   │   ├── test_text.csv               # 1,500 test text samples
│   │   ├── train_full.parquet          # Complete training records with embedded image bytes
│   │   ├── val_full.parquet            # Complete validation records with embedded image bytes
│   │   └── test_full.parquet           # Complete test records with embedded image bytes
│   └── dataset_card.md                 # Detailed MiRAGeNews provenance and schema card
├── notebooks/
│   ├── Environment_Setup.ipynb         # Step 1: GPU diagnostics and environment setup
│   ├── DataPreperations.ipynb          # Steps 2-4: Parquet inspection, cleaning, split, TF-IDF
│   └── training_Evaluation.ipynb       # Steps 5-11: Baselines, ResNet-18, fusion MLP, fine-tuning
├── models/
│   ├── tfidf_vectorizer.pkl            # Fitted 5,000-feature TF-IDF vectorizer artifact
│   ├── resnet_preprocess.pkl           # Torchvision image transform pipeline
│   ├── text_baseline_model.pkl         # Trained Logistic Regression baseline artifact
│   ├── veritasnet_best_model.pt        # Optimal Stage 1 weights (Val Acc: 85.87%)
│   └── veritasnet_finetuned_best.pt    # Optimal Stage 2 fine-tuned weights (Val Acc: 91.27%)
├── reports/
│   ├── experiment_log.md               # Step-by-step training metric log
│   ├── error_analysis.md               # Analysis of figurative & idiomatic headlines
│   ├── feature_reference.md            # Technical feature reference for oral examination
│   └── viva_notes.md                   # Concise viva reference notes
└── results/
    ├── final_experiment_comparison.csv # Baseline vs. Stage 1 vs. Stage 2 metric comparison
    ├── final_finetuned_test_confusion_matrix.png
    ├── text_baseline_multi_set_confusion_matrix.png
    └── epoch_vs_loss_curves.png
```

🚀 Reproduction & Execution Guide

Prerequisites

- Google Account with Google Drive access.
- Google Colab session with GPU acceleration enabled (**Runtime > Change runtime type > T4 GPU**).

Execution Sequence

1. Environment Diagnostics

Open and execute `notebooks/Environment_Setup.ipynb`:

- Mounts Google Drive at `/content/drive/MyDrive/VeritasNet`.
- Verifies Python 3.10+, PyTorch 2.0+, and NVIDIA Tesla T4 GPU allocation.
- Generates directory structure.

2. Data Preparation & Preprocessing

Open and execute `notebooks/DataPreperations.ipynb`:

- Loads `data/raw/mirage_news_raw.parquet` ($10,000$ instances).
- Performs MD5 image hashing and text sanitization ($9,997$ records retained).
- Executes stratified $70/15/15$ splitting with `random_state=42`.
- Fits `TfidfVectorizer(max_features=5000, ngram_range=(1,2))` on training headlines only.
- Serializes processed splits and vectorizer artifacts.

3. Training, Evaluation & Benchmarking

Open and execute `notebooks/training_Evaluation.ipynb`:

- **Cell 2:** Trains and evaluates the text-only Logistic Regression baseline (75.53% test accuracy).
- **Cell 3:** Precomputes 512-D visual feature vectors using pretrained ResNet-18 under `torch.no_grad()`.
- **Cell 4:** Initializes PyTorch multimodal DataLoaders.
- **Cell 5:** Instantiates the `VeritasNetFusion` network architecture.
- **Cell 6:** Executes Stage 1 training (10 epochs); saves best checkpoint at Epoch 2 (85.87% val accuracy).
- **Cell 7:** Evaluates Stage 1 model on the test set (86.27% test accuracy).
- **Cell 8:** Executes Stage 2 targeted fine-tuning on ResNet-18 `layer4` (3 epochs, differential learning rates).
- **Cell 9:** Evaluates champion model on unseen test set (91.20% test accuracy, 0.9118 Macro-F1).

📚 References & Academic Lineage

1. **Primary Benchmark:**
   Huang, R., Dugan, L., Yang, Y., & Callison-Burch, C. (2024). *MiRAGeNews: Multimodal Realistic AI-Generated News Detection*. Findings of the Association for Computational Linguistics: **EMNLP 2024**, pages 16436–16448. arXiv:2410.09045.
2. **Convolutional Visual Backbone:**
   He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), pages 770–778.
3. **Vision-Language Foundations:**
   Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. (2021). *Learning Transferable Visual Models From Natural Language Supervision*. Proceedings of the 38th International Conference on Machine Learning (**ICML**), PMLR 139, pages 8748–8763.
4. **Statistical Text Modeling:**
   Rajaraman, A., & Ullman, J. D. (2011). *Mining of Massive Datasets*. Cambridge University Press.
5. **Deep Learning Framework:**
   Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al. (2019). *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. Advances in Neural Information Processing Systems (**NeurIPS**) 32, pages 8024–8035.

📄 Citation

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  author={Sharma, Tanmay},
  journal={B.Tech Capstone Project Report},
  year={2025},
  institution={Department of Computer Science and Engineering}
}
```